



H2S

BHARATIYA ANTARIKSH HACKATHON 2026



Team Name : DigilIndia

Team Leader Name : Yashika Soni

Problem Statement : To develop and demonstrate an algorithm to predict energetic particle fluxes of electrons at geostationary orbit. The algorithm should be able to predict the harsh radiation fluxes at least 30 to 45 minutes in advance, and also give a reasonable forecast for 6 hours and 12 hours ahead.

Team Members

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Team Member-1:

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College: University College of Engineering
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KAVACH

How different is it from any of the other existing ideas?

- Existing/typical approaches treat this as a single black-box ML regression problem: solar wind in → one neural network → flux out, with one accuracy table at the end.
- KAVACH** instead runs two physically independent **forecasting engines in parallel**, a data-driven **Temporal Fusion Transformer** and a first-principles **radial-diffusion** physics solver and uses their agreement/disagreement as a real, physically grounded uncertainty signal, the way operational centers like **NOAA SWPC** reason about ensemble forecasts.
- It also explicitly solves the **data-scarcity problem** most teams will ignore: instead of training a weak model directly on **ISRO's** limited **1-2 years** of **GRASP/GSAT** data, it pretrains on **11 years** of **GOES** and transfers to the Indian-longitude payload turning a stated dataset limitation into a demonstrated engineering result.
- It includes a **regime-aware layer** (quiet / storm-onset / main phase / recovery) so the system adapts its behavior to the dominant physical process at each moment, rather than forcing one model to fit all conditions equally.

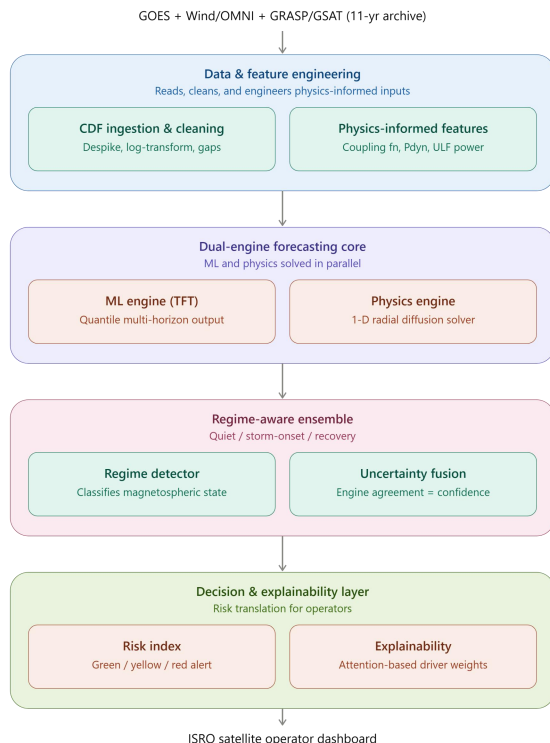
How will it be able to solve the problem?

- Reads and processes **11 years** of **GOES >2 MeV** electron flux and **Wind/OMNI** solar wind data from **CDF** archives; applies despiking, log-transformation, gap handling, and physics-informed feature engineering (**IMF** coupling function, dynamic pressure, **ULF** wave power proxy).
- The **dual-engine** ensemble produces multi-horizon forecasts simultaneously **30 minutes**, **6 hours**, and **12 hours** ahead exactly as required, with **quantile/probabilistic** outputs instead of single-point estimates.
- A **risk-translation layer** converts forecasted flux + uncertainty into a **Green/Yellow/Red** operational index based on known **deep-dielectric** charging thresholds, giving ISRO satellite operators a directly actionable output rather than a raw number.
- Validated** against real historical storm events (onset → peak → recovery) and cross-checked against **ISRO's** own **GRASP/GSAT** data for the Indian longitude sector.

USP of the proposed solution

- Hybrid physics + AI**, not AI alone and **defensible scientific grounding**, not just curve-fitting.
- Calibrated **uncertainty** as a feature, not an afterthought - operators get confidence, not just a number.
- Built-in domain transfer** to **ISRO's** own payload despite limited local data.
- Decision-ready output (risk index)** directly usable in satellite operations, not just a research metric.
- Explainable: **attention weights** tie forecast drivers back to known **radiation-belt** physics at each time horizon, directly demonstrating scientific understanding alongside technical execution.

List of features offered by KAVACH



01

Data ingestion & physics-informed feature engineering

- Automated reading and visualization of **GOES** electron flux and **Wind/OMNI** solar wind **CDF** archives (**11 years**)
- Built-in cleaning: **despiking**, **log-transformation**, **gap** interpolation
- Physics-derived features: **IMF** coupling function, dynamic **pressure**, **ULF** wave power proxy, lagged flux history

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Dual-engine forecasting core

- ML engine: Temporal Fusion Transformer producing **simultaneous** 30-min / 6-hr / 12-hr forecasts
- Physics engine: a numerically integrated **radial** diffusion solver, run independently as a first-principles cross-check
- Quantile/probabilistic outputs : **confidence** ranges, not single-point guesses

03

Regime-aware ensemble

- Automatic classification of **magnetospheric** state (quiet / storm-onset / main phase / recovery)
- Ensemble fusion** where ML-physics agreement or disagreement becomes a real, physically meaningful uncertainty signal

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Decision & explainability layer

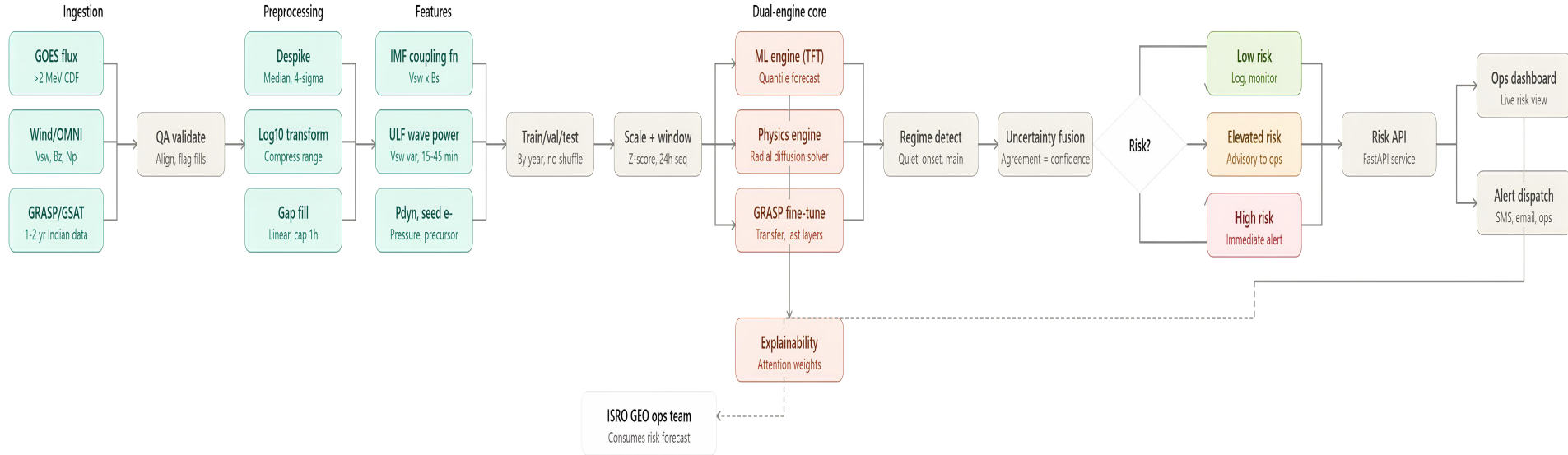
- Green/Yellow/Red operational **risk index** based on deep-dielectric charging thresholds
- Attention-based driver** explainability, showing which solar wind variable dominates at each forecast horizon

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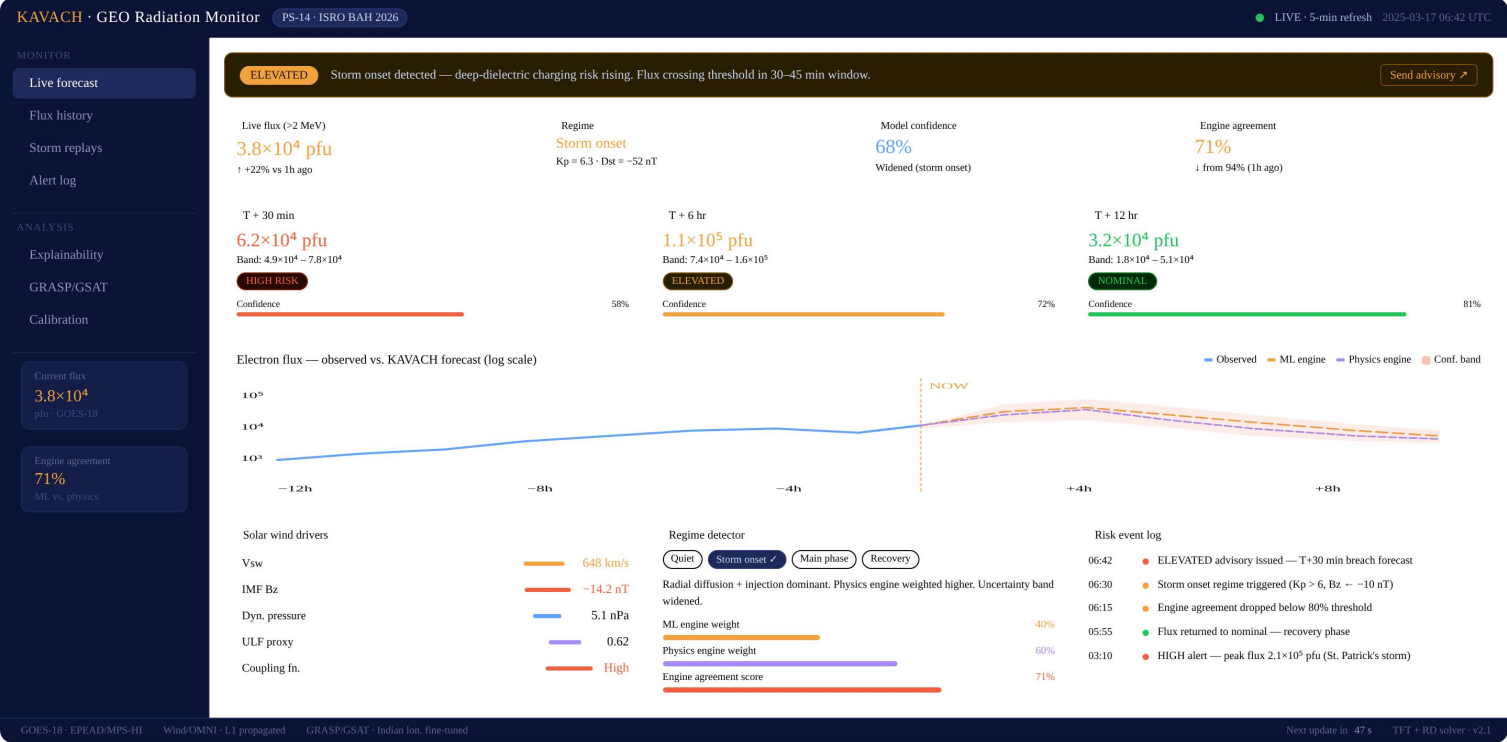
Cross-cutting capability

- Transfer learning from **GOES** (11 yrs) to **ISRO's GRASP/GSAT** data (1–2 yrs) for **Indian-longitude validation**
- Storm-replay validation** suite benchmarking against persistence and calibration plots

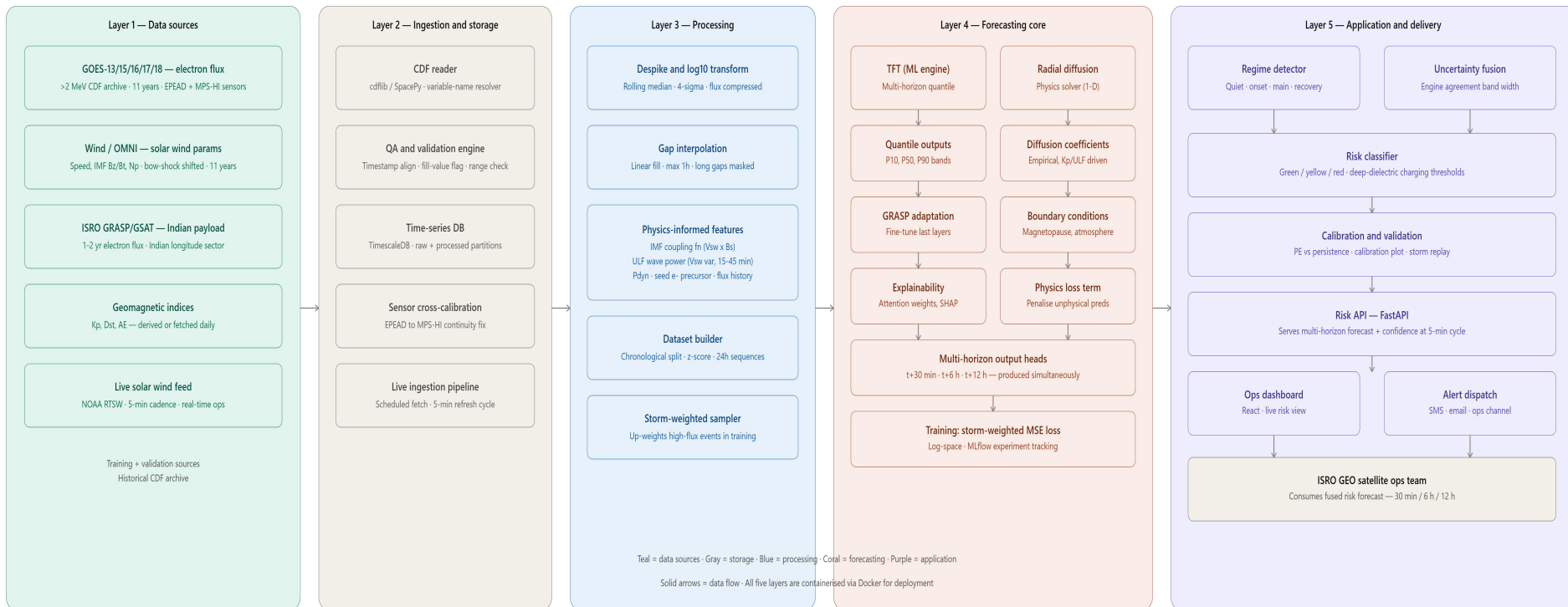
Process flow diagram



KAVACH dashboard mockup



Architecture diagram of the KAVACH



Technologies to be used in KAVACH :

Data ingestion & processing

- **Python** - primary language for the full pipeline
- **cdflib / SpacePy** - reading **GOES**, **Wind**, and **OMNI CDF** archive files
- **pandas, NumPy** - time-series alignment, resampling, cleaning
- **SciPy** - interpolation, signal filtering (despiking), and the numerical **ODE** solver for the radial diffusion physics engine

Forecasting core

- **PyTorch** - primary deep learning framework
- **PyTorch Forecasting / Darts** - Temporal Fusion Transformer implementation and multi-horizon quantile forecasting utilities
- **scikit-learn** - baseline models, scaling, evaluation metrics (**RMSE**, **PE**)
- **SHAP / built-in attention extraction** - explainability layer (driver importance per forecast horizon)

Validation & visualization

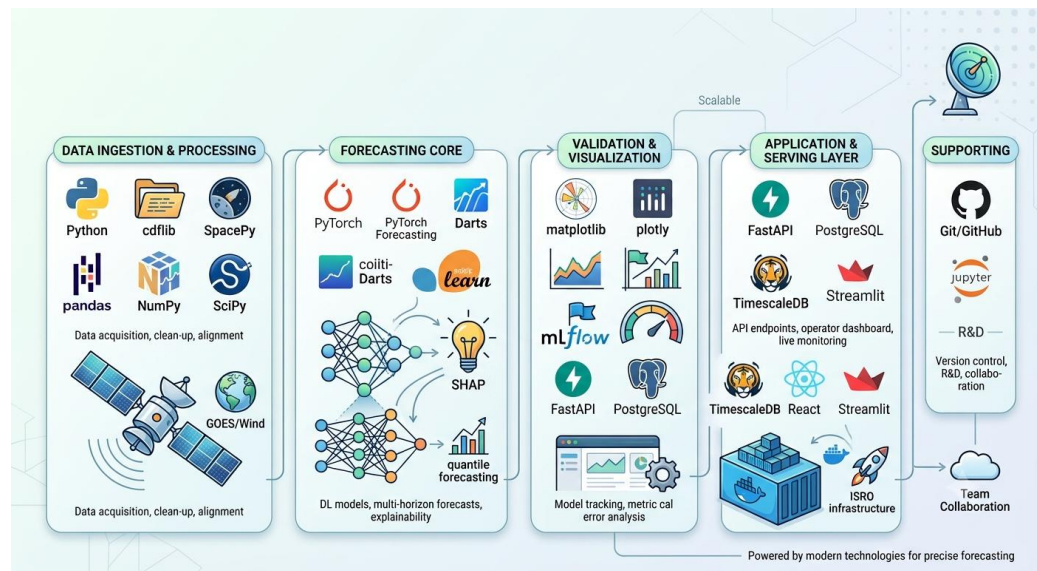
- **Matplotlib / Plotly** - flux time series, storm-replay plots, calibration curves
- **MLflow** (or **Weights & Biases**) - experiment tracking across model versions and hyperparameter runs

Application & serving layer

- **FastAPI** - Risk API serving forecasts and risk classifications
- **PostgreSQL / TimescaleDB** - time-series storage for processed flux, solar wind features, and forecast history
- **React** (or **Streamlit** for a faster build) - operator dashboard for live risk monitoring
- **Docker** - containerized deployment of the full pipeline for portability to **ISRO** infrastructure

Supporting

- **Git/GitHub** - version control and team collaboration
- **Jupyter** - exploratory analysis and model development during **R&D** phase



Estimated implementation cost :

Phase and Component	One-Time Cost	Recurring / yr
Phase 1 — Prototype (Hackathon stage)		
Open-source stack — Python, PyTorch, FastAPI, React	₹0	—
GOES / Wind / GRASP data — provided by problem statement	₹0	—
Google Colab / Kaggle GPU for training (free tier)	₹0	—
Team effort — student time, own laptops	₹0	—
Phase 2 — Pilot Validation (if ISRO takes it forward)		
Cloud GPU training — 3-4 hrs on A100, 10-15 runs	₹15,000 – 40,000	—
TimescaleDB / PostgreSQL setup — open source, VPS hosting	₹5,000 – 10,000	₹6,000 – 12,000
FastAPI + React dashboard — VPS hosting	₹5,000 – 8,000	₹12,000 – 20,000
1 integration engineer (part-time, 3 months)	₹60,000 – 90,000	—
Phase 3 — ISRO Operational Deployment (on existing ISRO HPC)		
Hardware — ISRO HPC already exists, marginal cost only	₹0 – 50,000	—
Integration into ISRO monitoring pipeline	₹1 – 2 lakh	—
Annual model retraining — cloud GPU, once per year	—	₹10,000 – 25,000
Part-time maintenance engineer	—	₹60,000 – 1.2 lakh
Total one-time (Phase 2 + 3)		
	₹1.85 – 3.5 lakh	—
Total recurring per year		
	—	₹88,000 – 1.57 lakh
Equivalent commercial space-weather subscription		
	—	₹40 – 80 lakh/yr

Net Saving to ISRO from year one: ₹38 - 78 Lakh

All software Open-Source + Zero licensing fees + inference runs on existing ISRO CPU infrastructure.

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THANK YOU